ScientoPy v 3.0.1, User Manual

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1 Installation

1.1 Pre-built releases (recommended)

1. Download the latest stable release for your platform from: `https://github.com/jpruiz84/ScientoPy/releases`

- **Windows:** Download `ScientoPy-windows-x64.zip`, unzip and run `ScientoPyGui.exe`
- **Linux:** Download `ScientoPy-linux-x64.tar.gz`, extract and run `./ScientoPyGui`
- **macOS:** Download `ScientoPy-macos-arm64.zip`, unzip and run `./ScientoPyGui`

Note (Windows): On first run, Windows SmartScreen may show a warning. Click "More info" then "Run anyway". This occurs because the application is not code-signed.

1.2 From source

1. Clone the repository: `git clone https://github.com/jpruiz84/ScientoPy.git`

2. Create a virtual environment and install dependencies:

```
cd ScientoPy
python -m venv venv
source venv/bin/activate   # Linux/macOS
venv\Scripts\activate      # Windows
pip install -r requirements.txt
```

3. Run the GUI: `python ScientoPyGui.py`

2 Interface overview

ScientoPy uses a PySide6 (Qt 6) graphical interface with four tabs, a menu bar, and a status bar. The window is resizable with a minimum size of 853×480 pixels.

2.1 Menu bar

- **View** → **Appearance:** Switch between Light, Dark, and System theme modes. The selected mode is saved and restored on next launch.

2.2 Keyboard shortcuts

- **Ctrl+O** – Browse and select a dataset folder
- **Ctrl+R** – Run (preprocess on Tab 1, analysis on Tab 2)
- **Ctrl+1/2/3/4** – Switch to the corresponding tab
- **Ctrl+C** – Copy selected cells in the Results or Extended Results tables

2.3 Status bar

The status bar at the bottom of the window shows feedback messages such as "Preprocessing complete – 5,548 papers", "Analysis complete – 10 topics found", or error information.

2.4 Configuration persistence

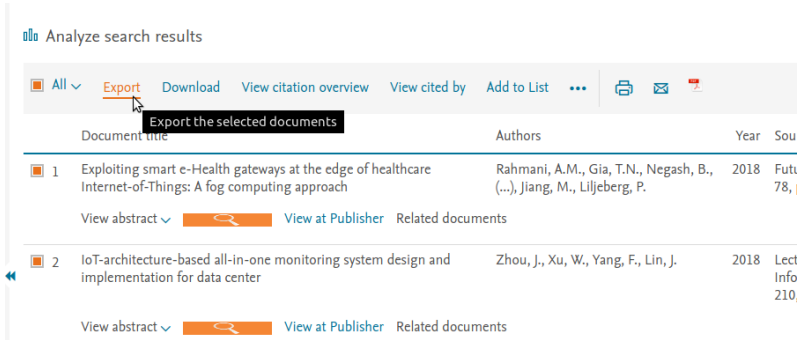
ScientoPy saves the following settings to `scientopy_config.json` and restores them on next launch: window size and position, splitter proportions, appearance mode, last dataset folder, and all analysis parameters (criterion, graph type, year range, topics length, skip first, window width, and checkbox states).

3 Download the bibliometric dataset

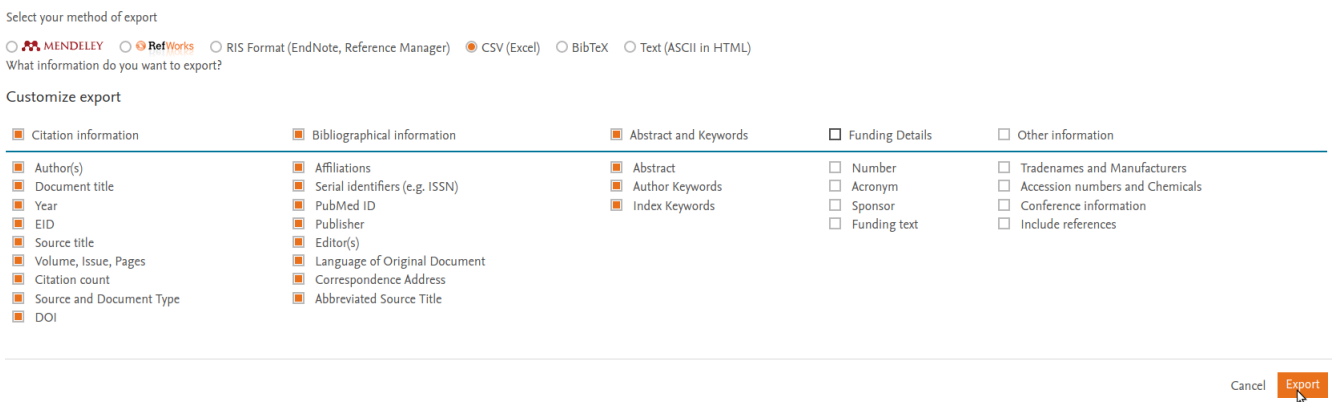
This section describes how to download the proper dataset from Scopus and WoS. Define a search criteria that will be used for Scopus and WoS. For this guide and for the example dataset we are using: "Bluetooth low energy".

3.1 Download the dataset from Scopus

1. Make your search with the defined search criteria for Article title, Abstract, Keywords.
2. Select all the results and click on Export:



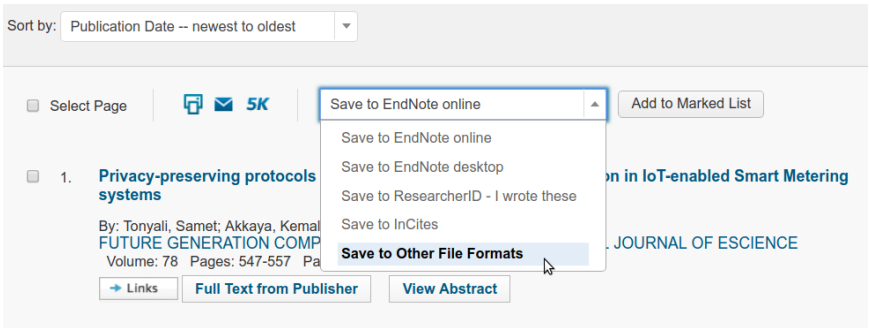
3. Select as method of export **CSV (Excel)**, and select the Customize export **Citation information, Bibliographical information, Abstract and Keywords**, then click on Export:



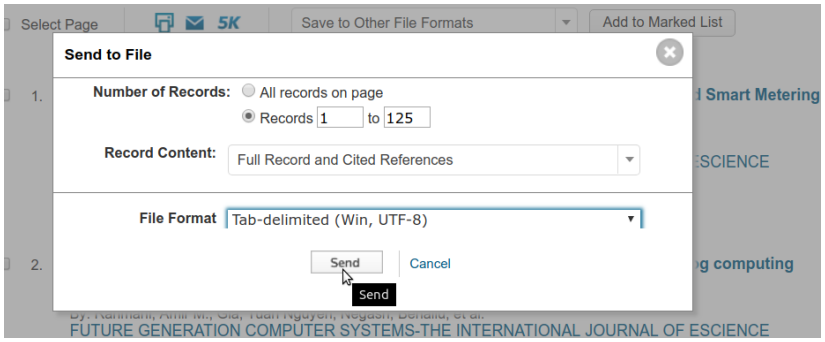
4. Save the file on the dataset folder (e.g. `/ScientoPy/dataIn`)

3.2 Download the dataset from WoS

1. Make your search with the defined search criteria for Topic.
2. Select **Save in Other File Formats**



3. Select the number of records to download, on Record Content select **Full Record and Cited References**, on File Format select **Tab-delimited (Win, UTF-8)**, and click on Send.



4. Save the file on the dataset folder (e.g. `/ScientoPy/dataIn`)

4 Preprocessing (Tab 1)

First we need to preprocess the downloaded dataset. This preprocess merges all the downloaded files from the dataset folder into a single file. Also, this process removes the duplicated documents. To preprocess the example dataset ("Bluetooth low energy" located in `dataInExample`), open ScientoPy and follow the next steps:

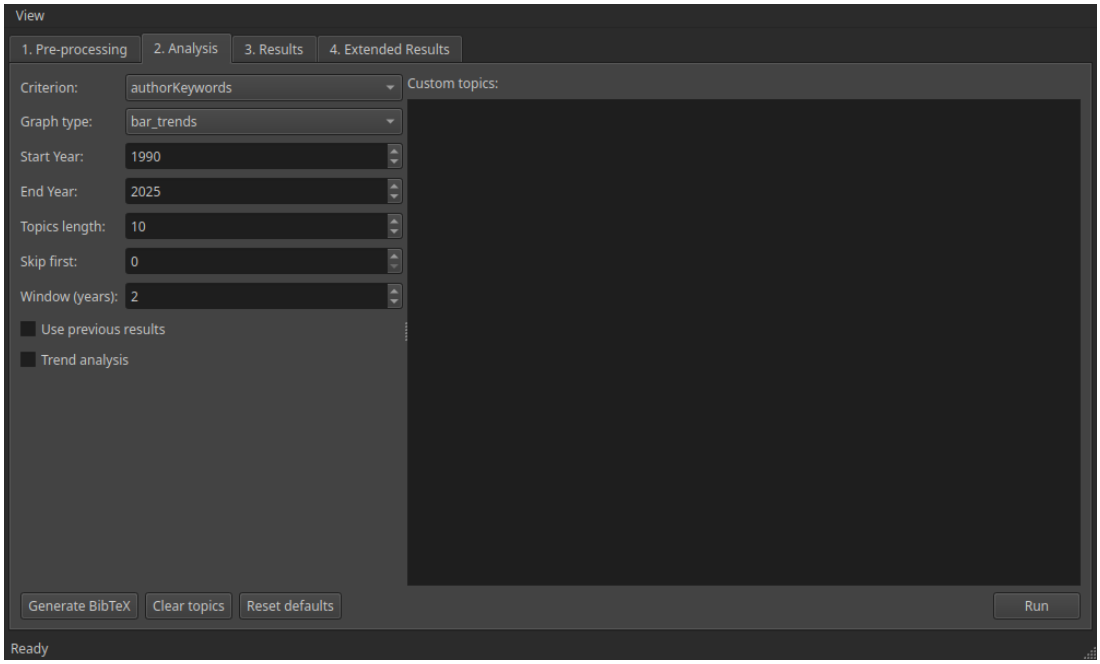
1. Click the `Browse` button, and select the folder that contains the downloaded dataset. For the example dataset, select the folder `dataInExample` inside the ScientoPy directory. Alternatively, use the keyboard shortcut **Ctrl+O**.
2. Ensure the `Remove duplicated documents` checkbox is checked (default).
3. Click the `Run preprocess` button (or press **Ctrl+R**). A progress dialog will appear showing the preprocessing status.
4. After preprocessing completes, the preprocess brief graph will appear and the application will automatically switch to the **3. Results** tab showing the preprocess brief table.

As result, inside the folder `ScientoPy/dataPre` you will find the following files:

- **papersPreprocessed.csv:** this file contains the information of all papers after the pre-process. This file will be used as input data for the analysis.
- **PreprocessedBrief.csv:** this file briefs the pre-process statistics results, such as duplicated papers removed, types of documents, and others.

5 Analysis (Tab 2)

The analysis tab lets you perform scientometric analysis operations. Click on the `2. Analysis` tab to access it.



5.1 Analysis parameters (left panel)

- **Criterion:** the search criterion to analyze, such as author keywords, author names, countries, etc. See Table 1 for the full list.
- **Graph type:** the type of output graph. See Section 10 for details.
- **Start year:** limit the start year for the analysis and graphs.
- **End year:** limit the end year for the analysis and graphs.
- **Topics length:** number of top topics to find in the selected criterion.
- **Skip first:** number of first top topics to skip during the analysis.
- **Window (years):** window width in years for growth indicators (AGR, ADY).
- **Use previous results:** run the analysis based on the output documents from the previous run.
- **Trend analysis:** find the top trending topics with the highest average growth rate.

5.2 Custom topics (right panel)

The right panel contains a text area for entering custom topics. Enter one topic per line. Topics can be grouped with commas and separated by line breaks. The asterisk (*) wildcard is supported. See Section 5.6 for details.

5.3 Action buttons (bottom bar)

- **Generate BibTeX:** generate a BibTeX bibliography file from a LaTeX document (see Section 8).
- **Clear topics:** clear the Custom topics text area.
- **Reset defaults:** reset all analysis parameters to their default values (criterion: authorKeywords, graph type: bar_trends, years: 1990 to last year, topics length: 10, skip first: 0, window: 2, checkboxes unchecked, custom topics cleared).
- **Run:** execute the analysis with the current parameters. Can also be triggered with **Ctrl+R**.

After the analysis completes, results are displayed in the **3. Results** and **4. Extended Results** tabs, and the output graph appears.

5.4 Criterion descriptions

The ScientoPy criterion are described in Table 1:

Table 1: ScientoPy criterion description

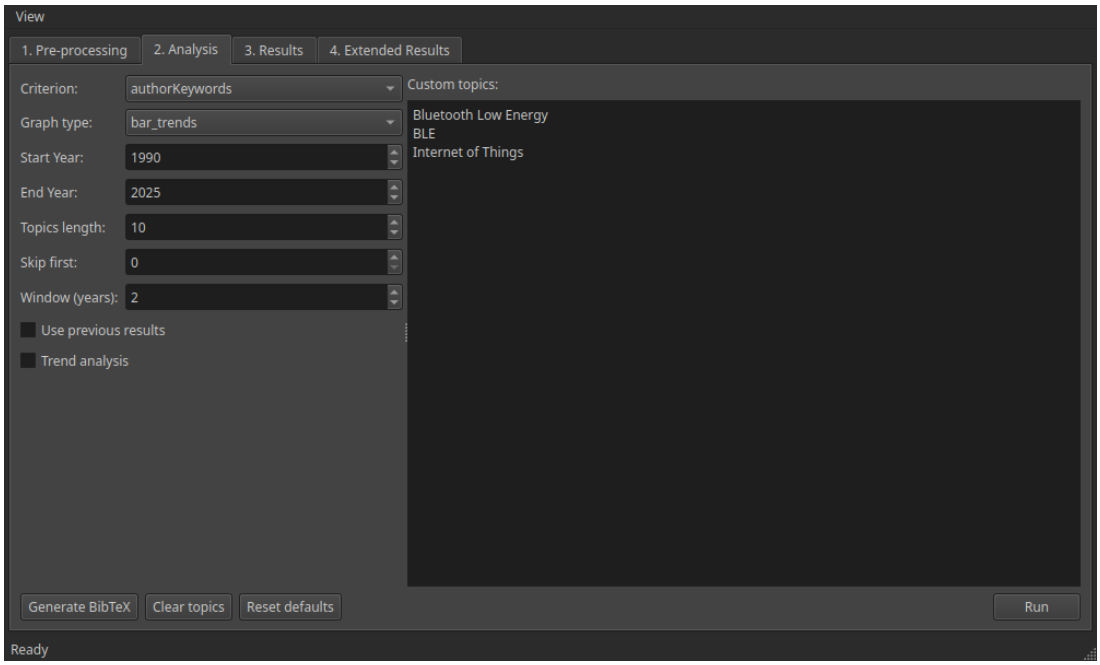
Criterion	Description
author	Authors last name and first name initial
sourceTitle	Publication or journal name
subject	Research areas, only from WoS documents
authorKeywords	Author keywords
indexKeywords	Keywords generated by the index, from WoS *Keyword Plus*, and from Scopus *Indexed keywords*
bothKeywords	AuthorKeywords and indexKeywords are used for this search
abstract	Document abstract, for use with pre-defined topics and asterisk wildcard
documentType	Type of document
dataBase	Database where the document was extracted (WoS or Scopus)
country	Country extracted from authors affiliations
institution	Institution extracted from authors affiliations
institutionWithCountry	Institution with country extracted from authors affiliations

5.5 Extract top topics

To find the top 10 author keywords, keep the default parameters in the analysis tab and click the `Run` button. This will generate a list with the top 10 topics on the selected criterion (in this case authorKeywords), with the number of documents per topic. The bar trends graph will appear, and the quantitative results are saved in the folder `ScientoPy/results` and displayed in the **3. Results** tab.

5.6 Analyze custom topics inside a criterion

If you want to analyze custom topics, such as comparing two countries' publication trends, enter the topics in the `Custom topics` text area (one topic per line):



You can analyze any topic in any criterion. You can also group two or more topics into one by separating them with commas. This is useful for abbreviations and plural/singular forms, for example:

```
WSN, Wireless sensor network, Wireless sensor networks
RFID, RADIO FREQUENCY IDENTIFICATION
```

Tip: You can also use the `Analyze selection` button in the **3. Results** tab to copy selected topic names directly into the Custom topics field (see Section 6).

5.7 Asterisk (*) wildcard

You can use the asterisk wildcard to find phrases or words which start or end with the entered text. For example, entering `device*` in the Custom topics area will match "device", "devices", "device integration", etc.

ScientoPy prints the topics found for the wildcard-based search in the console output. You can copy this information to analyze each specific topic found individually.

5.8 Evolution plot

Select `evolution` as the Graph type to see an evolution plot. This option plots the accumulative documents, average documents per year (ADY), and percentage of documents in the last years (PDLY).

Note: select the right start and end year limits that fit your plot lines best.

5.9 Finding trending topics

ScientoPy can find the top trending topics based on the highest average growth rate (AGR). The AGR is calculated using the following equation:

$$AGR = \frac{\sum_{i=Y_s}^{Y_e} P_i - P_{i-1}}{(Y_e - Y_s) + 1},$$

where:

AGR = Average growth rate;
Y_s = Start year;
Y_e = End year;
P_i = Number of publications on year i.

To find the top trending topics, check the `Trend analysis` checkbox and click `Run`. ScientoPy will find the top 200 topics, calculate the AGR for the specified window period, and sort them from highest to lowest AGR.

5.10 Analysis based on the previous results

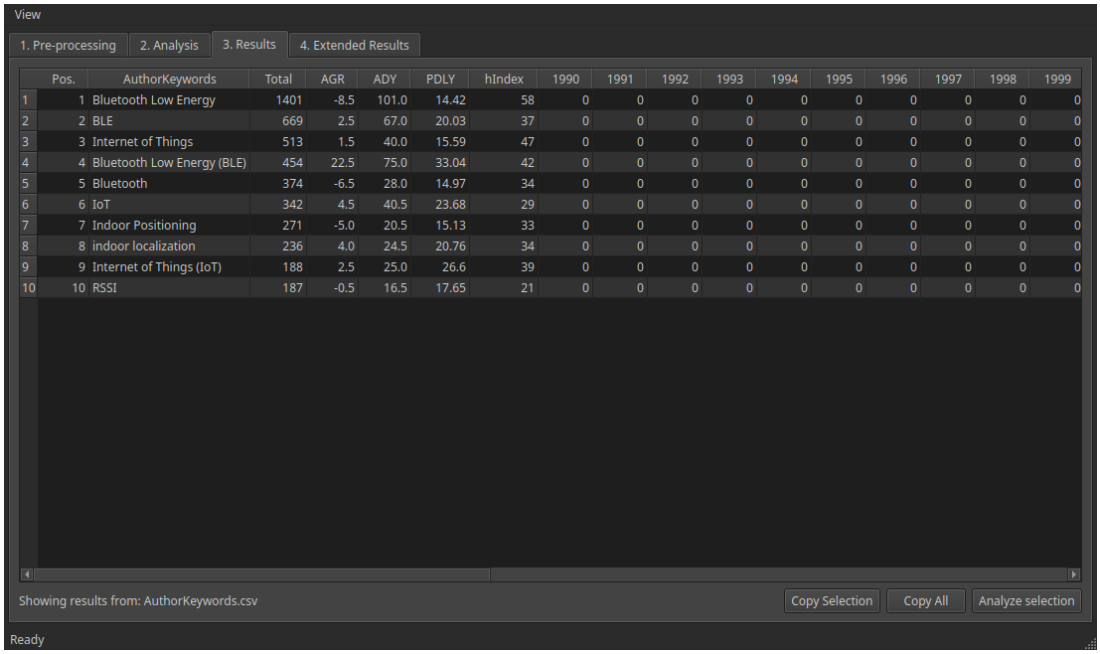
ScientoPy generates an output file with all the output documents from the last run. For example, if you run an analysis with criterion `country` and custom topic `Canada`, ScientoPy creates a documents output file (`results/papersPreprocessed.csv`) with all documents that have authors with affiliation in Canada.

This output file can be used as input for a subsequent analysis by checking the `Use previous results` checkbox. For example, you could then select criterion `authorKeywords` to find the top author keywords from Canadian papers, or select criterion `country` to find which countries have the most co-authored documents with Canada.

Note: the output documents file is only generated when `Use previous results` is **not** checked. Running multiple analyses with this option checked will not overwrite the documents output file.

6 Results tab (Tab 3)

The Results tab displays the analysis output in a sortable table. After running a preprocess, it shows the preprocessing brief. After running an analysis, it shows the top topics with their statistics (Total, AGR, ADY, PDLY, h-index, and documents per year).



View

1. Pre-processing | 2. Analysis | 3. Results | 4. Extended Results

	Pos.	AuthorKeywords	Total	AGR	ADY	PDLY	hIndex	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999
1	1	Bluetooth Low Energy	1401	-8.5	101.0	14.42	58	0	0	0	0	0	0	0	0	0	0
2	2	BLE	669	2.5	67.0	20.03	37	0	0	0	0	0	0	0	0	0	0
3	3	Internet of Things	513	1.5	40.0	15.59	47	0	0	0	0	0	0	0	0	0	0
4	4	Bluetooth Low Energy (BLE)	454	22.5	75.0	33.04	42	0	0	0	0	0	0	0	0	0	0
5	5	Bluetooth	374	-6.5	28.0	14.97	34	0	0	0	0	0	0	0	0	0	0
6	6	IoT	342	4.5	40.5	23.68	29	0	0	0	0	0	0	0	0	0	0
7	7	Indoor Positioning	271	-5.0	20.5	15.13	33	0	0	0	0	0	0	0	0	0	0
8	8	indoor localization	236	4.0	24.5	20.76	34	0	0	0	0	0	0	0	0	0	0
9	9	Internet of Things (IoT)	188	2.5	25.0	26.6	39	0	0	0	0	0	0	0	0	0	0
10	10	RSSI	187	-0.5	16.5	17.65	21	0	0	0	0	0	0	0	0	0	0

Showing results from: AuthorKeywords.csv

Copy Selection | Copy All | Analyze selection

Ready

6.1 Column tooltips

Hover over column headers to see tooltips with definitions of abbreviations:

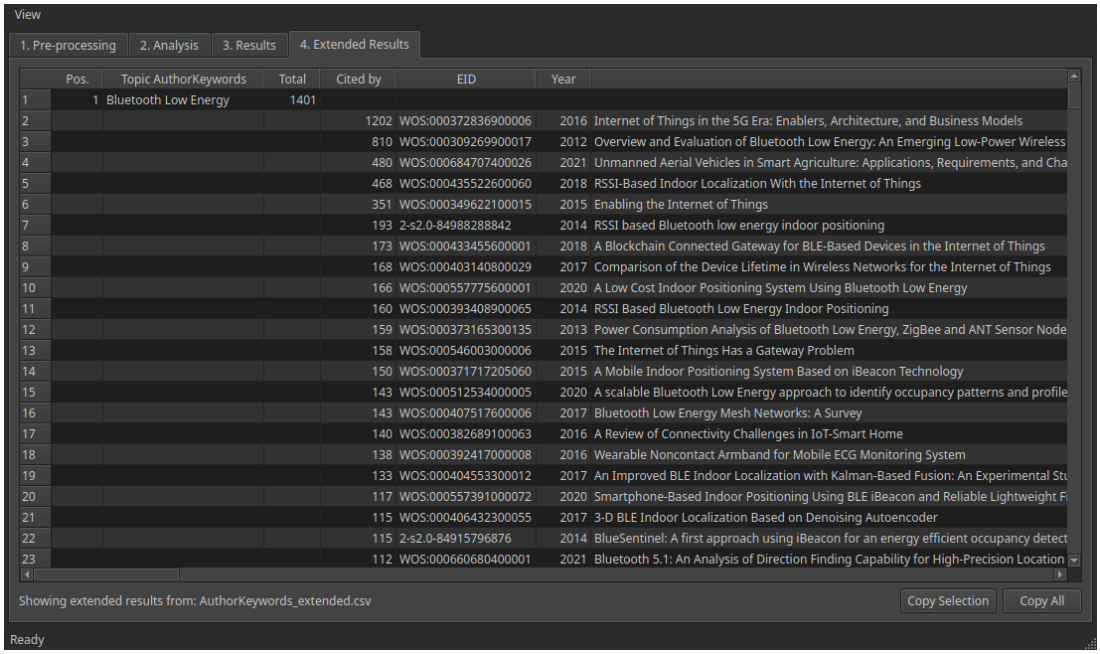
- **AGR:** Average Growth Rate – mean year-over-year change in publications within the window period.
- **ADY:** Average Documents per Year – mean publications per year within the window period.
- **PDLY:** Percentage of Documents in Last Years – proportion of total documents published within the window period.
- **hIndex:** h-index – the largest number h such that h documents have at least h citations each.

6.2 Action buttons

- **Copy Selection:** copy the selected cells to the clipboard (also available via Ctrl+C).
- **Copy All:** copy the entire table to the clipboard.
- **Analyze selection:** copy the topic names from the selected rows to the `Custom topics` text area in the Analysis tab, and switch to the Analysis tab. This is useful for quickly selecting topics from the results for further analysis with a different graph type or criterion.

7 Extended Results tab (Tab 4)

The Extended Results tab shows the detailed per-document output after running an analysis. Each row represents a paper associated with one of the analyzed topics, including authors, title, year, source, citations, DOI, and other metadata.



View

1. Pre-processing | 2. Analysis | 3. Results | 4. Extended Results

	Pos.	Topic AuthorKeywords	Total	Cited by	EID	Year	
1	1	Bluetooth Low Energy	1401				
2				1202	WOS:000372836900006	2016	Internet of Things in the 5G Era: Enablers, Architecture, and Business Models
3				810	WOS:000309269900017	2012	Overview and Evaluation of Bluetooth Low Energy: An Emerging Low-Power Wireless
4				480	WOS:000684707400026	2021	Unmanned Aerial Vehicles in Smart Agriculture: Applications, Requirements, and Cha
5				468	WOS:000435522600060	2018	RSSI-Based Indoor Localization With the Internet of Things
6				351	WOS:000349622100015	2015	Enabling the Internet of Things
7				193	2-s2.0-84988288842	2014	RSSI based Bluetooth low energy indoor positioning
8				173	WOS:000433455600001	2018	A Blockchain Connected Gateway for BLE-Based Devices in the Internet of Things
9				168	WOS:000403140800029	2017	Comparison of the Device Lifetime in Wireless Networks for the Internet of Things
10				166	WOS:000557775600001	2020	A Low Cost Indoor Positioning System Using Bluetooth Low Energy
11				160	WOS:000393408900065	2014	RSSI Based Bluetooth Low Energy Indoor Positioning
12				159	WOS:000373165300135	2013	Power Consumption Analysis of Bluetooth Low Energy, ZigBee and ANT Sensor Node
13				158	WOS:000546003000006	2015	The Internet of Things Has a Gateway Problem
14				150	WOS:000371717205060	2015	A Mobile Indoor Positioning System Based on iBeacon Technology
15				143	WOS:000512534000005	2020	A scalable Bluetooth Low Energy approach to identify occupancy patterns and profile
16				143	WOS:000407517600006	2017	Bluetooth Low Energy Mesh Networks: A Survey
17				140	WOS:000382689100063	2016	A Review of Connectivity Challenges in IoT-Smart Home
18				138	WOS:000392417000008	2016	Wearable Noncontact Armband for Mobile ECG Monitoring System
19				133	WOS:000404553300012	2017	An Improved BLE Indoor Localization with Kalman-Based Fusion: An Experimental Stu
20				117	WOS:000557391000072	2020	Smartphone-Based Indoor Positioning Using BLE iBeacon and Reliable Lightweight F
21				115	WOS:000406432300055	2017	3-D BLE Indoor Localization Based on Denoising Autoencoder
22				115	2-s2.0-84915796876	2014	BlueSentinel: A first approach using iBeacon for an energy efficient occupancy detect
23				112	WOS:000660680400001	2021	Bluetooth 5.1: An Analysis of Direction Finding Capability for High-Precision Location

Showing extended results from: AuthorKeywords_extended.csv

Copy Selection | Copy All

Ready

The table is sortable by clicking column headers and supports cell selection with Ctrl+C copy.

7.1 Action buttons

- **Copy Selection:** copy the selected cells to the clipboard.
- **Copy All:** copy the entire table to the clipboard.

8 Generate BibTeX

With ScientoPy you can generate a LaTeX BibTeX file for your references. Follow the next steps:

1. Run an analysis to generate results.
2. Open the Extended Results tab (Tab 4) and find the EID values of the papers you want to cite.
3. In your LaTeX document, use the paper EID values as citation keys inside `\cite{}` commands, for example: `\cite{2-s2.0-85012345678}`
4. Save your LaTeX document.
5. In the Analysis tab, click the `Generate BibTeX` button and select the LaTeX document.
6. ScientoPy will find all EID citation keys in the document and generate a BibTeX file with the corresponding bibliographic entries. The output file is saved next to the input file as `<filename>_bibliography.bib`.

Note: for your reference we have included an article LaTeX file example in `latexExample/article_example.tex`. This contains cites from the example dataset. To test the Generate BibTeX capability:

- Preprocess the example dataset located in `dataInExample`
- In the Analysis tab, click `Generate BibTeX`
- Select the file `latexExample/article_example.tex`
- Compile the LaTeX file with: `latexmk -pdf article_example.tex`

9 Output files and directories

After running ScientoPy analysis you will find the following folder and files structure:

```
ScientoPy
├── dataInExample
├── dataPre
│   ├── papersPreprocessed.csv
│   └── PreprocessedBrief.csv
├── graphs
├── latexExample
├── Manual
├── results
│   ├── AuthorKeywords.csv
│   ├── AuthorKeywords_extended.csv
│   └── papersPreprocessed.csv
└── scientopy_config.json
```

These folders and output files are described below:

- **dataInExample:** contains Scopus and WoS example data set for the search criteria "Bluetooth low energy" downloaded on April 2, 2026. This is the input example for preprocessing.
- **dataPre:** output folder for the preprocessing results, and input folder for the analysis.
- **papersPreprocessed.csv:** preprocessed papers data with all input documents merged, filtered, and duplicates removed. This is the input file that the analysis uses.
- **PreprocessedBrief.csv:** preprocessing brief table that shows the preprocessing results related to total papers found per database, the omitted papers, the duplicated papers count per database, and the total number of papers per paper type (Conference paper, article, review...).
- **graphs:** graphs output folder for preprocessing and analysis.
- **latexExample:** folder with the LaTeX file article example to generate the BibTeX from the preprocessed dataset.
- **Manual:** folder with the PDF manual.
- **results:** output folder for analysis result files.
- **AuthorKeywords.csv:** analysis output file for the selected criterion (in this case authorKeywords) that shows the top or custom topics with the total number of documents, the Average Growth Rate (AGR), the Average Documents per Year (ADY), the Percentage of Documents in Last Years (PDLY), the h-index, and the documents per each year.
- **AuthorKeywords_extended.csv:** analysis output file for the selected criterion that shows the top or custom topics with the documents related to each one.
- **papersPreprocessed.csv:** inside the results folder, this file contains the output papers from the last analysis run. This is used as input when the option `Use previous results` is checked.
- **scientopy_config.json:** user configuration file storing window geometry, appearance mode, and analysis parameters. Automatically created and updated.

10 ScientoPy graph types

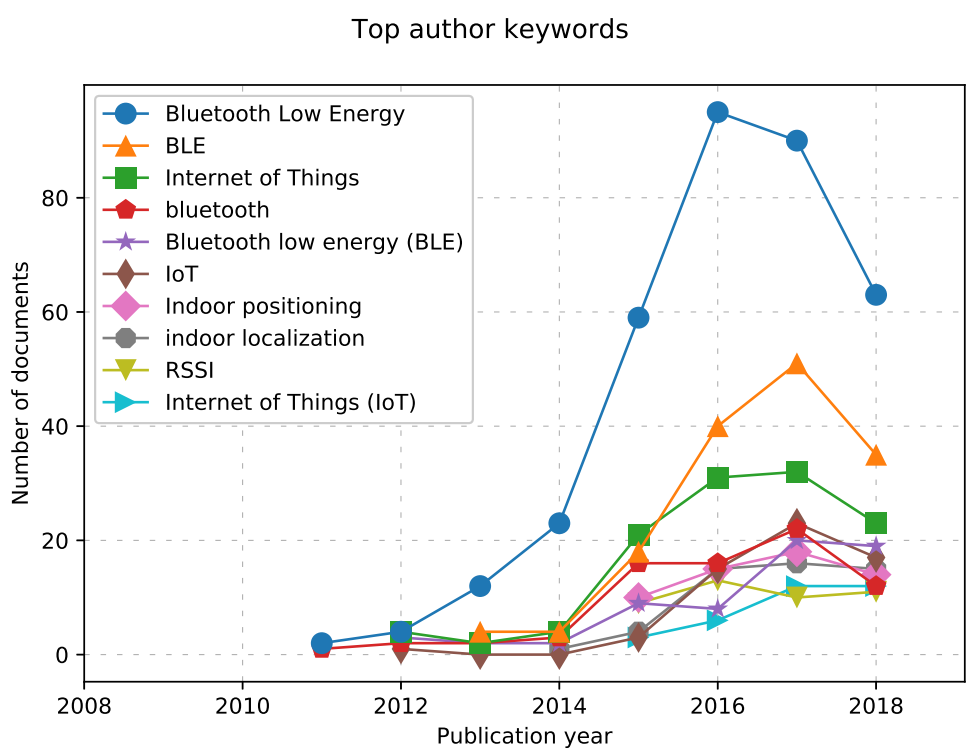
ScientoPy has 5 different ways to graph the results described on Table 2.

Table 2: ScientoPy output graph types

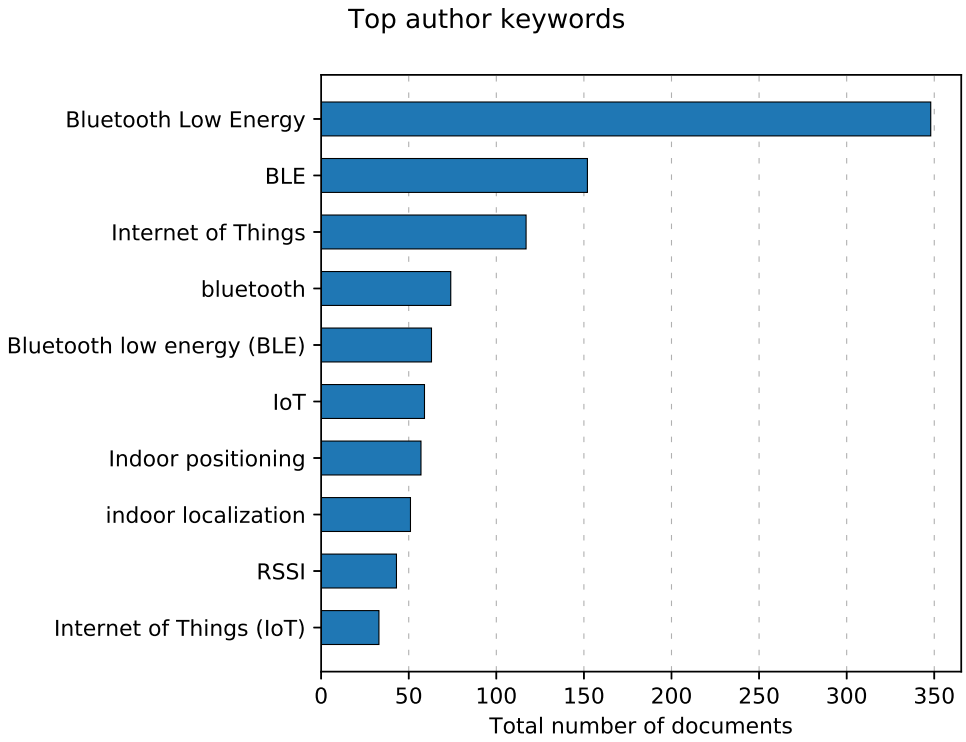
Graph type	Abbreviation	Description
Time line	`time_line`	Graphs the number of documents of each topic vs the publication year
Horizontal bars	`bar`	Graphs the total number of documents of each topic in horizontal bars
Horizontal bars trends	`bar_trends`	Graphs the total number of documents of each topic in horizontal bars, with the percentage of documents published in the last years
Evolution	`evolution`	Graphs two plots, one with the accumulative number of documents vs the publication year, and another with the average papers per year vs the percentage of documents in the last years
Word cloud	`word_cloud`	Generate a word cloud based on the topic total number of publications

Below are some examples of these graph types.

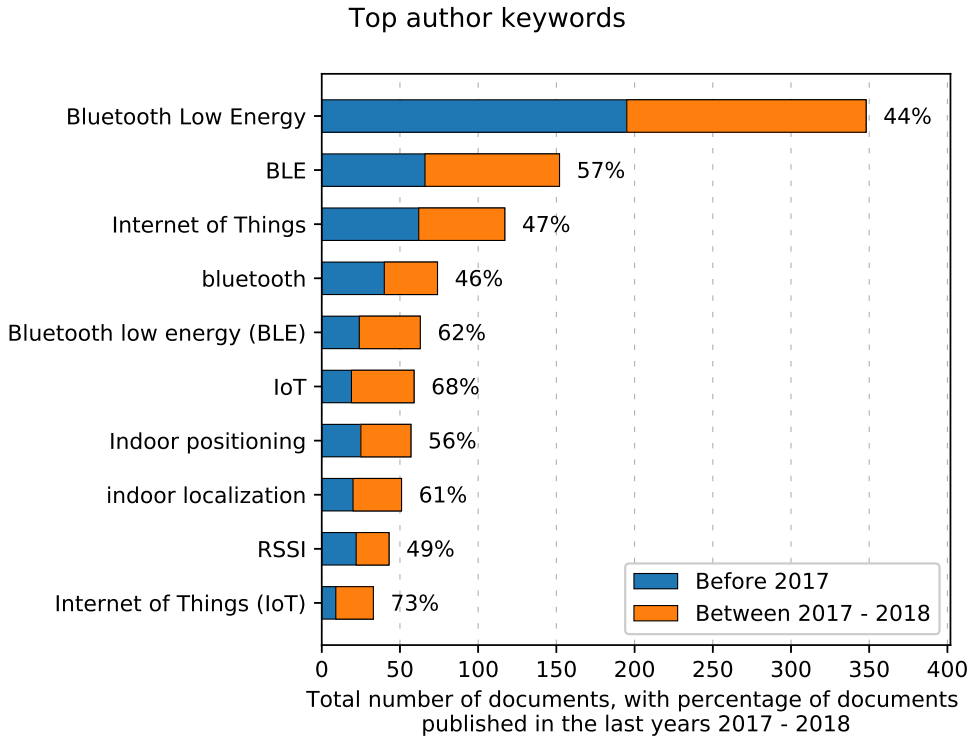
10.1 Time line graph



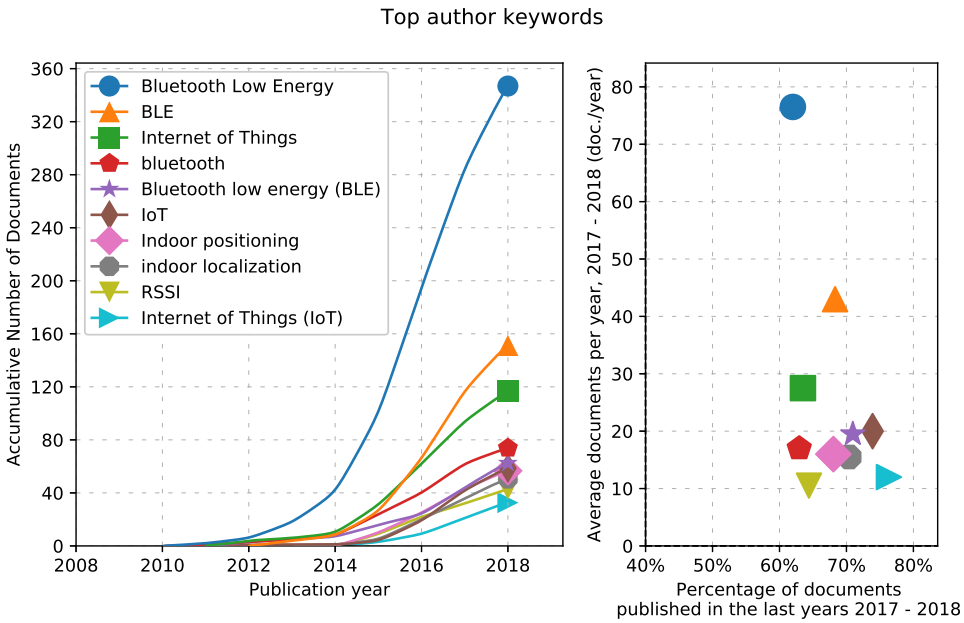
10.2 Horizontal bars graph



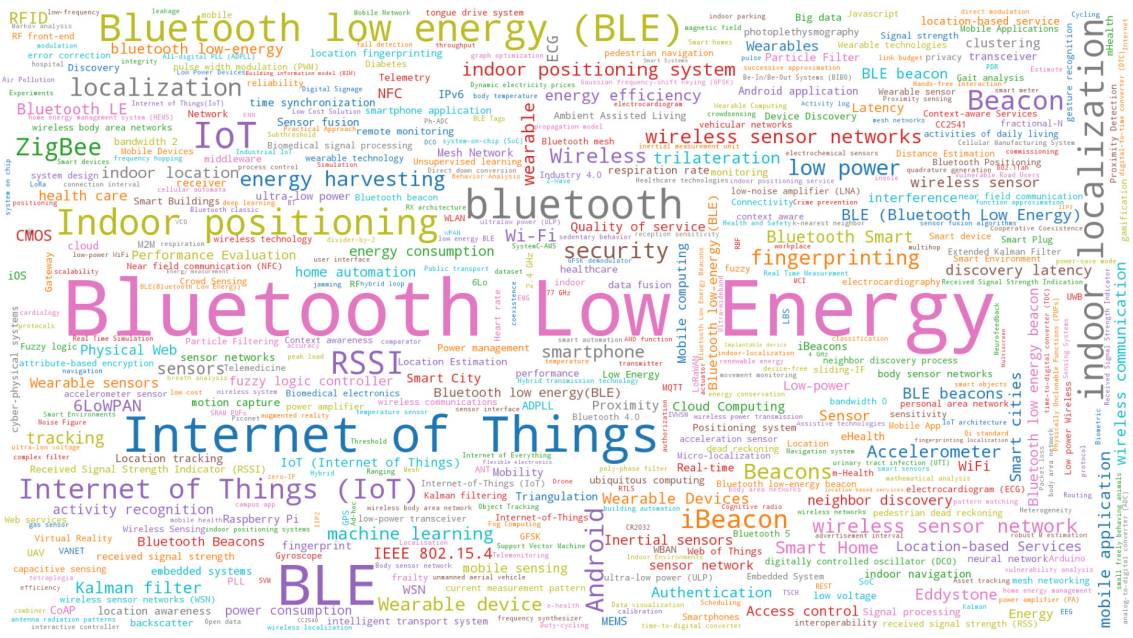
10.3 Horizontal bars trends



10.4 Evolution graph



10.5 Word cloud graph


Bluetooth low energy (BLE)
Bluetooth Low Energy
Internet of Things
Internet of Things (IoT)
indoor positioning system
localization
IoT
ZigBee
Indoor positioning
bluetooth
wireless sensor networks
Wireless trilateration
energy harvesting
Beacon
Wearables
clustering
BLE beacon
Android application
Latency
low power
interference
BLE (Bluetooth Low Energy)
Bluetooth Smart
fingerprinting
discovery latency
security
smartphone
RSSI
Wearable sensors
fuzzy logic controller
Smart City
Bluetooth low energy(BLE)
Cloud Computing
Sensor
BLE beacons
Accelerometer
Beacons
Wearable Devices
neighbor discovery
iBeacon
wireless sensor network
Inertial sensors
Smart Home
Location-based Services
machine learning
Bluetooth Beacons
IEEE 802.15.4
mobile sensing
BLE
Wearable device
Kalman filter
Authentication
Eddystone
Access control
Android
indoor
localization
Wireless communication
mobile application
Smart cities
Bluetooth low energy beacon
Mobile computing
Bluetooth low-energy
Physical Web
6LoWPAN
tracking
activity recognition
Raspberry Pi
Gyroscope
embedded systems
CoAP
location awareness
power consumption
backscatter
intelligent transport system
Smartphones
MEMS
Energy
Signal processing
received signal strength (RSS)
interoperability
Wi-Fi
CMOS
health care
home automation
energy consumption
Performance Evaluation
Near field communication (NFC)
Crowd Sensing
Smart Buildings
Quality of service
Smart Environment
Extended Kalman Filter
Location tracking
Received Signal Strength Indicator (RSSI)
IoT (Internet of Things)
Internet-of-Things (IoT)
Triangulation
Kalman filtering
Mobility
ubiquitous computing
Real-time
WiFi
Inertial sensors
digitally controlled oscillator (DCO)
neural network
Arduino
sensor network
Embedded System
indoor navigation
mesh networking
Asset tracking
Proximity Detection
Cooperative Coexistence
Extended Kalman Filter
Smart Plug
Smart device
Smart Environment
Received Signal Strength Indication
Wireless communication
Bluetooth low energy beacon
Smart cities
mobile application
Energy
EEG
Cyber-physical systems
Gamification
Digital-to-time converter (DTC)
Internet
mHealth
Proximity Detection
Gesture recognition
Cycling
Location-based service
Mobile Applications
Signal strength
photoplethysmography
Big data
Javascript
Wearable technologies
Particle Filter
transceiver
Wearable sensor
Context-aware Services
Distance Estimation
activities of daily living
Bluetooth Positioning
Cellular Manufacturing System
Bluetooth mesh
Industry 4.0
Healthcare technologies
Wireless sensor
Bluetooth Low energy(BLE)
Quality of service
Smart Environment
Extended Kalman Filter
Smart device
Smart Plug
Cooperative Coexistence